



University of Asia Pacific

Department of Computer Science And Engineering

LAB REPORT

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Submitted By :

Name : Mashrur Rahman Masfi

Registration No : 23101182

Section : D (D1)

Submitted To :

Atia Rahman Orthi

Lecturer

Dept. of CSE

FCFS Disk Scheduling Algorithm

1. Problem Statement:

Given Parameters:

- **Request Queue:** [55, 58, 39, 18, 90, 160, 150, 38, 184]
- **Initial Head Position:** 100
- **Scheduling Policy:** FCFS

Execution Sequence: Initiating at cylinder 100, the read/write head simply follows the queue's natural order. The traversal path is as follows: **100 → 55 → 58 → 39 → 18 → 90 → 160 → 150 → 38 → 184**

Mathematical Calculation: The total seek time (head movement) is computed by summing the absolute differences between consecutive cylinder positions: Total Movement = $|100 - 55| + |55 - 58| + |58 - 39| + |39 - 18| + |18 - 90| + |90 - 160| + |160 - 150| + |150 - 38| + |38 - 184|$ Total Movement = 45 + 3 + 19 + 21 + 72 + 70 + 10 + 112 + 146 **Total Head Movement = 498 cylinder**

2. Source Code:

```
1 def fcfs_scheduling(requests, initial_head):
2
3     total_seek_time = 0
4     current_position = initial_head
5     seek_sequence = [current_position]
6
7     print(f"Initial Head Position: {current_position}")
8     print("Processing Requests...")
9
10    for request in requests:
11        # Calculate distance and add to total seek time
12        distance = abs(current_position - request)
13        total_seek_time += distance
14
15        # Move head to the new request position
16        current_position = request
17        seek_sequence.append(current_position)
18
19    print(f"\nSeek Sequence: {' -> '.join(map(str, seek_sequence))}")
20    print(f"Total Head Movement: {total_seek_time} cylinders")
21
22    return total_seek_time
23
24 request_queue = [55, 58, 39, 18, 90, 160, 150, 38, 184]
25 head_start = 100
26
27 fcfs_scheduling(request_queue, head_start)
```

3. Screenshot of output:

```
Initial Head Position: 100
Processing Requests...

Seek Sequence: 100 → 55 → 58 → 39 → 18 → 90 → 160 → 150 → 38 → 184
Total Head Movement: 498 cylinders
```

Scan (Elevator) Algorithm

Given Parameters:

- Request Queue: [55, 58, 39, 18, 90, 160, 150, 38, 184]
- Initial Head Position: 100
- Disk Boundaries: 0 (Left) and 200 (Right)

Case A: Left-Direction SCAN

Seek Sequence: 100 → 90 → 58 → 55 → 39 → 38 → 18 → 0 → 184

Mathematical Calculation: Total Movement = $|100 - 90| + |90 - 58| + |58 - 55| + |55 - 39| + |39 - 38| + |38 - 18| + |18 - 0| + |0 - 184|$ Total Movement = $10 + 32 + 3 + 16 + 1 + 20 + 18 + 184$ Total Head Movement = 284 cylinders

Source Code :

```
1 def fcfs_scheduling(requests, initial_head):
2
3     total_seek_time = 0
4     current_position = initial_head
5     seek_sequence = [current_position]
6
7     print(f"Initial Head Position: {current_position}")
8     print("Processing Requests...")
9
10    for request in requests:
11        # Calculate distance and add to total seek time
12        distance = abs(current_position - request)
13        total_seek_time += distance
14
15        # Move head to the new request position
16        current_position = request
17        seek_sequence.append(current_position)
18
19    print(f"\nSeek Sequence: {' -> '.join(map(str, seek_sequence))}")
20    print(f"Total Head Movement: {total_seek_time} cylinders")
21
22    return total_seek_time
23
24 request_queue = [55, 58, 39, 18, 90, 160, 150, 38, 184]
25 head_start = 100
26
27 fcfs_scheduling(request_queue, head_start)
```

ScreenShot:

```
Left SCAN Seek Sequence: 100 -> 90 -> 58 -> 55 -> 39 -> 38 -> 18 -> 0 -> 184  
Total Head Movement: 284 cylinders
```

Right SCAN:

Seek Sequence: 100 → 150 → 160 → 184 → 200

Mathematical Calculation: Total Movement = $|100 - 150| + |150 - 160| + |160 - 184| + |184 - 200|$ Total Movement = 50 + 10 + 24 + 16 **Total Head Movement = 100 cylinders**

Source Code :

```
1 def scan_right_scheduling(requests, initial_head, disk_end=200):
2
3     right_requests = sorted([r for r in requests if r >= initial_head])
4
5
6     seek_sequence = [initial_head] + right_requests + [disk_end]
7
8
9     total_seek_time = sum(abs(seek_sequence[i] - seek_sequence[i+1])
10                          for i in range(len(seek_sequence) - 1))
11
12     print(f"Right SCAN Seek Sequence: {' -> '.join(map(str, seek_sequence))}")
13     print(f"Total Head Movement: {total_seek_time} cylinders")
14     return total_seek_time
15
16
17 request_queue = [55, 58, 39, 18, 90, 160, 150, 38, 184]
18 head_start = 100
19
20
21 scan_right_scheduling(request_queue, head_start, disk_end=200)
```

Screenshot:

```
Right SCAN Seek Sequence: 100 -> 150 -> 160 -> 184 -> 200
Total Head Movement: 100 cylinders
```

4. Conclusion:

This laboratory experiment successfully demonstrated the practical implementation and performance analysis of two fundamental disk scheduling algorithms: FCFS and SCAN. The primary objective was to evaluate how varying scheduling logic impacts the physical movement of the disk arm and overall system efficiency.

Key Observations:

FCFS Inefficiency: As expected, the FCFS algorithm yielded the highest total head movement (498 cylinders). Because it strictly follows the arrival order without considering the physical location of the requests, it causes excessive back-and-forth traversal.

SCAN Optimization: The SCAN algorithm significantly optimized the seek time by processing requests directionally. The Right-direction SCAN was the most efficient for this

specific dataset, resulting in a minimal head movement of only 100 cylinders. The Left-direction SCAN also performed substantially better than FCFS, requiring 284 cylinders.

The empirical data conclusively proves that directional scheduling algorithms like SCAN are vastly superior to non-directional algorithms like FCFS. By minimizing unnecessary cylinder traversal, SCAN reduces mechanical wear on the disk drive and lowers the average response time for I/O requests. Furthermore, this lab reinforced our ability to translate theoretical OS concepts into functional Python logic, bridging the gap between academic theory and practical software engineering

GitHub Link: <https://github.com/MasfiRahman/CSE-402-Operating-System-Lab-.git>